DASH Engine Design

Sophia Klymchuk, Adin Sacho-Tanzer, Devin Zhang
ME-433: Rocket Science
Prof. Fabian Chacon
December 19, 2025

Contents

Introduction

In this project, we designed and analyzed the performance of a simple ramjet engine. The main design goal was to achieve positive thrust for operation at 50,000 ft between Mach 2.75 and Mach 3.25. This ramjet is called the DASH engine, after the names of the designers: Devin, Adin, and SopHia.

Ramjets operate on the Brayton cycle. The three basic components of a Brayton cycle are compression, heat addition, and expansion. Unlike other jet engines, the ramjet accomplishes the compression and expansion processes without any moving parts, instead using carefully selected geometry changes and the principles of supersonic aerodynamics and compressible flow.

Because the ramjet operates on a Brayton cycle, it can be characterized using the known formulas of Brayton cycles. Specifically, the efficiency of a Brayton cycle

$$\eta = 1 - \frac{1}{P_r^{(\gamma-1)/\gamma}}$$

where γ is the ratio of specific heats of the working fluid, and P_r is the pressure ratio of the cycle, which is the ratio of the pressure during the heat addition stage to the input/output pressure. Thus, to achieve adequate performance given that our input conditions were fixed as atmospheric, we needed to increase the pressure in the combustor as much as possible.

The ramjet we designed consists of multiple components, each of which performs a specific function and was analyzed individually. These components are listed in Table 1, with full analysis later in this report.

Table 1: Ramjet components and functions.

Component	**Function**
Inlet	Takes in supersonic flow, compresses it, and slows it to subsonic
Diffuser	Takes in subsonic flow, further compresses and slows it
Flameholder	Holds flame to cause combustion, leads to pressure drop
Combustor	Burns fuel, decreases static pressure, increases stagnation temperature
Converging section	Increases flow velocity to sonic
Nozzle	Expands flow to uniform conditions, pressure match atmosphere, increases velocity

The thrust of the ramjet was calculated using a control volume approach. A control volume was designated around the ramjet, and a momentum balance analysis was performed on the boundary to find the force of thrust. This analysis can be justified using Reynold's transport theorem for momentum:

$$\Sigma\vec{F} = \int_{CS} \rho\vec{V}\left(\vec{V}\cdot\hat{n}\right)dA$$

In addition, because the control volume experiences pressure forces, these forces must also be included in the analysis. However, this analysis can be simplified by cleverly choosing the control volume such that its surface is always normal to flow wherever flow exists. If this is done, the equation for thrust can be reduced to the following:

$$F = \Sigma F_{P,x} - \Sigma\rho_{in}V_{in}^2A_{in,x} + \Sigma\rho_{out}V_{out}^2A_{out,x}$$

where $F_{P,x}$ are pressure forces in the x-direction, ρ is density, V is velocity, and A is area. Thus, in order to fully calculate thrust for a ramjet of known geometry, we need to know pressure at all forward- and backward- facing faces and density and velocity at all inlets.

Component Analysis and Design

Assumptions

To simplify this analysis, we made multiple assumptions throughout. We assumed the gas throughout the engine consists of air, neglecting any composition or mass changes caused by fuel injection and burning. We modeled air as an ideal, calorically perfect gas throughout the engine, meaning that we assume constant properties of $c_p = 1005 \frac{\mathrm{J}}{\mathrm{kg\ K}}$, $\gamma = 1.4$, and $R = 287.05 \frac{\mathrm{J}}{\mathrm{kg\ K}}$. Finally, we assumed that all flow was steady and inviscid. We thus neglected all viscous drag effects, boundary layer effects, and any flow separation.

Inlet

Analysis

The purpose of the inlet in our ramjet is to slow and compress atmospheric flow for use in the later components. The inlet consists of a series of straight planes, each inclined upwards relative to the previous. As the incoming flow encounters each successive plane, it turns it upwards by the relative angle between the two planes and causes an oblique shock. This oblique shock compresses, slows, and turns the flow.

The flow characteristics throughout the inlet can be calculated using the theta-beta-mach relation, which relates the turning angle of an oblique shock (theta) to the angle of the shock itself (beta) to the incoming Mach number for the shock. We can then use other relations to relate pressures, temperatures, and densities before and after the shock. By doing this successively for each shock, we can calculate the properties at every point throughout the inlet, as seen in Figure 1. To find a 1-dimensional profile of these properties, we perform an average weighted by mass flux over each vertical slice of the 2-dimensional field.

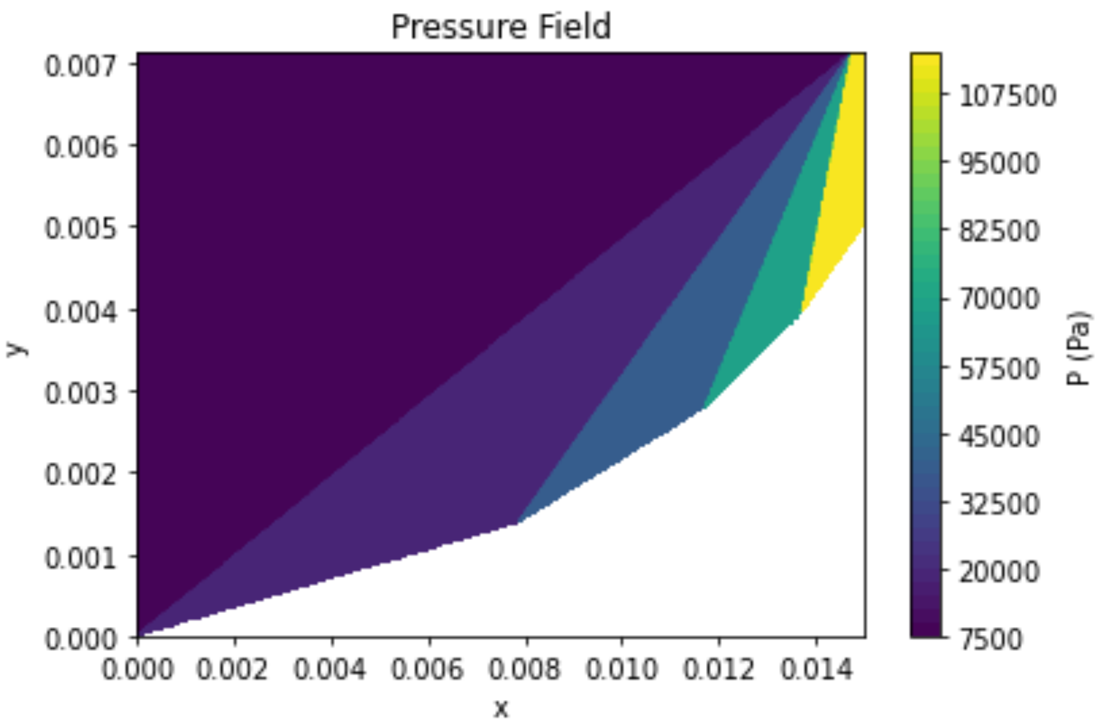

Figure 1: Pressure field throughout inlet

Finally, after the series of oblique shocks, a single normal shock exists which makes the flow subsonic. The properties of the flow after this shock can be calculated from the properties of the flow before this shock using the standard normal shock equations. Because this shock exists at the very end of the inlet, its properties cannot be seen in Figure 1; however, when calculating the properties of the flow received by the diffuser, the effects of the normal shock are accounted for.

Using the oblique shock calculations mentioned earlier, inlet drag can also be derived. Pressure drag on each face can be calculated as $F = P * l * w \sin\theta$, where P is the pressure on the face, l is the length of the face, w is the width of the face, and θ is the angle of the face relative to the horizontal.

Design

The design requirement for the inlet was for all shocks to converge on a single point at the very edge of the nozzle, what we called the “lip”. This is advantageous because the shocks compress air, so having shocks end at a point before the lip means that air is compressed which then does not pass through the jet. This leads to increased pressure on upper faces outside the jet, which causes increased pressure drag. On the other hand, if oblique shocks were to enter the inlet, past the lip, this would cause instabilities, which would have extremely negative consequences. Therefore, the inlet was designed such that at the maximum design Mach number, at which the shocks are steepest, all oblique shocks meet exactly at the lip, as seen in Figure 2.

This yields maximum effectiveness at the maximum design Mach, with effectiveness slightly falling as Mach number decreases, but without causing the inlet to swallow shocks.

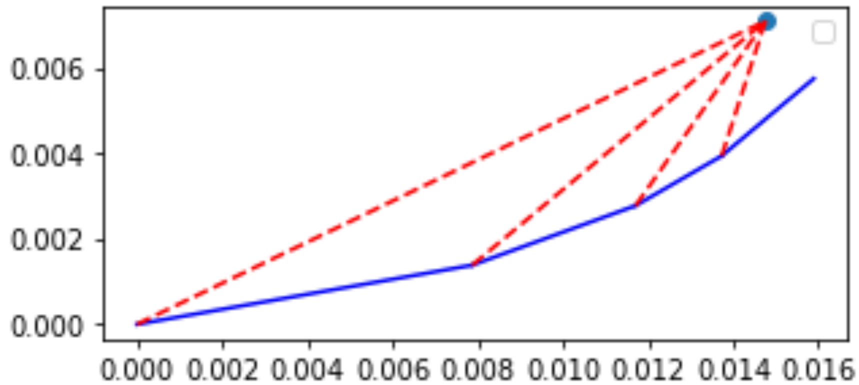

Figure 2: Diagram of inlet shocks

In order to get the exact shape of the inlet, two other parameters were necessary: the required mass flow rate of the inlet ($\dot{m}$), and the turning angles of the inlet planes. $\dot{m}$ determines the height of the inlet because the inlet effectively captures all air in front of it, between the front of the inlet and the lip. For given Mach number, atmospheric conditions, and engine width, a specific height is required to yield a desired mass flow rate. Once this is done and a series of turning angles for the inlet planes are determined, the design is constrained. If the first angle is arbitrarily placed at x=0, y=0, the intersection between its shock and the top of the inlet determines the location of the lip. Then, each subsequent turn can be placed by requiring its shock to intersect the lip.

In addition to the inlet, the lip needs some structure to connect it to the top wall. This was designed as follows: the last inlet surface was continued until it reaches its closest point to the lip. Because flow over that surface is parallel to the surface, the line between the surface and the lip at that point is orthogonal to the flow. Then, the flow turns horizontal in order to meet the diffuser. This is done by keeping the flow channel constant-area, so the top surface at this point must be a circle centered at the end of the inlet surface and with radius equal to the distance between that point and the lip. This circle continues until horizontal, at which point it can join the top wall of the rest of the ramjet. This surface gives us the inside surface of the top side of the ramjet.

The outside of the top surface is designed as follows. Whenever the conditions are such that the oblique shocks do not perfectly meet at the lip, they will all pass the lip on the left, so the flow over the lip will be the same as the flow after all inlet oblique shocks, but before the inlet normal shock. In order to minimize drag, the lip outer surface should be parallel to this flow in order to not cause an additional oblique shock, which would increase pressure even further. Therefore, the outer surface of the lip is parallel to the last section of the inlet. It has length sufficient to give the lip some thickness and then flattens out to form the top surface of the top wall. Note that this design does have a disadvantage for the case where all shocks do meet exactly at the lip because then flow over this surface will be completely horizontal, and thus a oblique shock will form in order to turn the flow to go along the surface. This issue is unavoidable, however, due to geometry constraints.

The overall lip design is shown in green in Figure 3, in combination with the end of the inlet surfaces in blue and the oblique shocks in red.

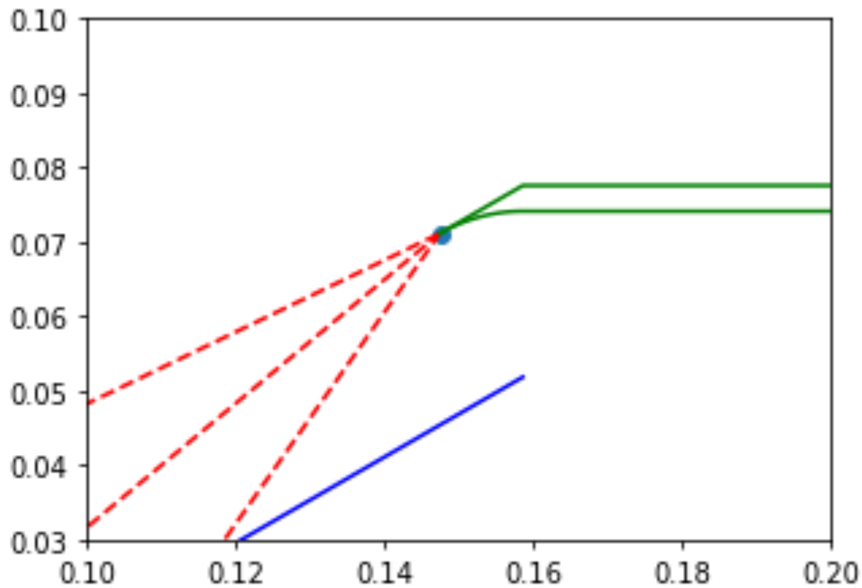

Figure 3: Lip design

The design in Figure 1, Figure 2 and Figure 3 are arbitrary designs with four turns, each 10 degrees. The parameters for the final inlet design can be found in Table 2.

Diffuser

The diffuser is treated as quasi-one-dimensional flow, which is assumed to be isentropic. The purpose of the diffuser is to decelerate incoming subsonic flow while maximizing pressure recovery and avoiding shocks. The diffuser is treated as a variable-area duct with constant stagnation properties.

Stagnation properties within the diffuser were modeled with Equations 1a and 1b, in which the dimensionless ratio of stagnation properties to static properties are functions of the Mach number M and the ratio of specific heats γ.

$$\frac{P_0}{P} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma - 1}} \tag{1a}$$

$$\frac{T_0}{T} = \left(1 + \frac{\gamma - 1}{2} M^2\right) \tag{1b}$$

Throughout the diffuser geometry, flow conditions are determined by the Area-Mach relation, which relates the area to the critical sonic area $A*$ as a function of M and γ.

$$\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2}\left[\frac{2}{\gamma + 1}\left(1 + \frac{\gamma - 1}{2} M^2\right)\right]^{\frac{\gamma+1}{\gamma-1}} \tag{2}$$

Design

The diffuser geometry is generated by enforcing a controlled reduction in Mach number from inlet to exit values. Incoming values of mass-flow rate, Mach, pressure, and temperature are taken from the outputs of the design's previous stage, whereas exit Mach number and diffuser length are design parameters that we chose.

Firstly, inlet area is calculated to ensure mass-flow continuity, where the speed of sound a_{in} is a function of temperature.

$$A_{in} = \frac{\dot{m}}{\rho_{in} u_{in}} = \frac{\dot{m}}{\left(\frac{P_{in}}{RT_{in}}\right)(M_{in} a_{in})} \tag{3a}$$

$$a_{in} = \sqrt{\gamma R T_{in}} \tag{3b}$$

Using inlet conditions, the stagnation pressure P_0, stagnation temperature T_0, and critical area A^* are computed. These quantities remain constant throughout the diffuser. The Mach number is enforced to decrease linearly from M_{in} to M_{exit} along the diffuser length. This choice provides a smooth, monotonic deceleration.

For each Mach number, we iterate through a list of x-values along the diffuser length. At each x-value, the required area is computed from the Area-Mach relation (Equation 2), the height is obtained from the fixed engine width into the page, and static pressure and temperature are recovered from isentropic relations (Equation 1). The result is a complete distribution of diffuser geometry and flow properties.

Evaluation

A secondary function is provided to evaluate flow properties for a given diffuser geometry defined by x-coordinates and wall height. Given inlet conditions, the inlet stagnation properties and critical area are computed. At each x-value, the local Mach number is obtained by inverting the area-Mach relation and solving numerically (via bisection method) for the subsonic solution. Static pressure and temperature are then recovered isentropically. This function is later used for evaluation of the complete engine design, given arbitrary (off-design) flow conditions.

Flameholder

The flameholder is an intermediate component between the diffuser and the combustor. Its function is to cause pockets of turbulence in the stream and ignite the fuel in those pockets. This holds the flame and allows it to continue burning without being extinguished.

In our analysis, we did not analyze any or design any physical footprint for the flameholder. Instead, we simulated it as an instantaneous pressure drop in accordance with the formula $\Delta P/P = 0.81\gamma M^2$, where P is pressure, γ is the ratio of specific heats of the fluid, and M is the incoming Mach number of the flow. This pressure drop exists because the flameholder consists of multiple small physical obstructions to the flow, which combine to cause a pressure drop.

Combustor

The combustor is modeled as a constant-area heat-addition process using Rayleigh flow relations (assumed to be frictionless). Under this framework, fuel addition increases the stagnation temperature while modifying the local Mach number and static properties.

We assume that the fuel is hydrogen with a lower heating value of $LHV_{H2} = 120 \ \frac{\text{MJ}}{\text{kg}}$, and that the introduction of fuel only contributes heat to our design (ignoring all associated mass and composition changes). Combustion completion is assessed using an empirical burn time correlation based on average pressure and stagnation temperature.

The combustor cross-sectional area is computed from continuity using inlet density and velocity, as displayed in Equation 3. The height h is held constant along the combustor, and is found by dividing the cross-sectional area by the into-the-page depth.

Rayleigh flow is conveniently expressed relative to the Rayleigh critical (star) state. The code computes inlet stagnation temperature and then the corresponding star properties:

$$\frac{P_0}{{P_0}^*} = \frac{1+\gamma}{1+\gamma M^2}\left(\frac{2+(\gamma-1)M^2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \tag{4a}$$

$$\frac{T_0}{{T_0}^*} = \frac{(\gamma+1)M^2}{(1+\gamma M^2)^2}\left(2+(\gamma-1)M^2\right) \tag{4b}$$

These star quantities refer provide the reference needed to map between stagnation temperature rise and Mach number change under Rayleigh flow.

Design

When a fuel mass flow is specified, the code determines how long the combustor must be to allow the reaction to complete (and for all our fluid to combust). Firstly, we note that the total heat release rate (in units of power) is:

$$\dot{Q} = \dot{m}_{H2} LHV_{H2} \tag{5a}$$

Thus, the specific heat addition per unit mass of air is:

$$q = \frac{\dot{Q}}{\dot{m}_{air}} \tag{5b}$$

Utilizing this, we can find our rise in stagnation temperature over the full combustor as:

$$T_{0,out} = T_{0,in} + \frac{q}{c_p} \tag{5c}$$

This value is used to numerically solve for an exit Mach number, enforcing a subsonic solution upon Equation 4b. With the exit Mach number known, we can find all static properties from stagnation. The combustor length is set using a residence-time requirement based on an empirical burn-time model. Firstly, the average velocity, pressure, and stagnation temperature throughout the combustor are found. Then, we utilize the correlation shown in Equation 6 to find burn time (in milliseconds).

$$\tau = 325P^{-1.6}e^{\frac{-0.8T_0}{1000}} \tag{6}$$

The resulting length is then simply the product between average velocity throughout combustor, u_{avg} and burn time (converted to seconds) τ.

Analysis

To evaluate the combustor for any given input, note that the combustor feeds directly into the converging section and nozzle. When these two later sections are operating off-design, they must still choke the flow at the throat (the narrowest part) of the converging section. A more detailed description is given below. However, for purposes of combustor design, the Mach number leaving the combustor must be the same, regardless of operating Mach.

Luckily, fuel flow is physically an easy parameter to control. Real engines can implement real-time control of how much fuel is provided. Thus, we can prescribe a different fuel mass-flow rate for each operating condition, such that our combustor produces the same exit Mach number regardless of the flow conditions coming in.

Thus, to analyze the combustor off-design, we firstly identify the on-design exit Mach number. Given this value, Equation 4b can be solved to find the required rise in stagnation temperature, which then provides a required amount of heat addition via Equation 5c, which can then be used to find the required amount of fuel flow via Equation 5a. After this is found, we utilize the burn time correlation of Equation 6 to simply check if the time spent flowing through the combustor is greater than or equal to the time it takes for the fuel to burn. This ensures that we input all necessary heat (though no more, as it is limited by a single fuel flow rate, as to not choke our flow).

Converging section

Design

The purpose of the converging section of a ramjet is to accelerate the subsonic flow coming from the combustor to sonic before entering the nozzle. During the design phase of this project, the shape of the converging section was left variable such that the flow reaches $M = 1$ at the minimum cross-section regardless of the conditions at the converging section's inlet.

The stagnation properties at the converging section's inlet were calculated from static properties using the isentropic relations shown in Equations 1a and 1b. These stagnation properties are assumed to remain constant throughout the converging section as the flow is assumed to be quasi-1D and isentropic.

The converging section's inlet area was then calculated based on the mass flow rate continuity and the inlet conditions as seen in Equation 3a. To ensure the flow reaches sonic at the throat, the Area-Mach relation, seen in Equation 2, was used to calculate A*, the sonic throat area, for the calculated inlet properties. The throat conditions were then calculated using the same isentropic relations shown in Equations 1a and 1b, assuming stagnation pressure and temperature is conserved through the converging section.

As a result of this assumption, the exit conditions are only dependent on the inlet properties and the exit Mach, which is set to M=1. Therefore, the geometric contour between the inlet and the throat is unconstrained. A cubic polynomial, shown in Equation 7, was used to design the bottom surface of the contour.

$$A(\xi) = A_{in} - (A_{in} - A_{throat})(3\xi^2 - 2\xi^3), \qquad where\ \xi = \frac{x}{L} \tag{7}$$

This ensures that $\frac{dA}{dx} = 0$ at both boundaries to make sure the transitions between sections are smooth. In the real world, this would help minimize any flow separation and pressure losses that would occur from sharp turns but does not impact the analysis for this project. The top surface of the contour was left flat for simplicity. An example of a converging section contour with arbitrary input conditions is shown in Figure 4.

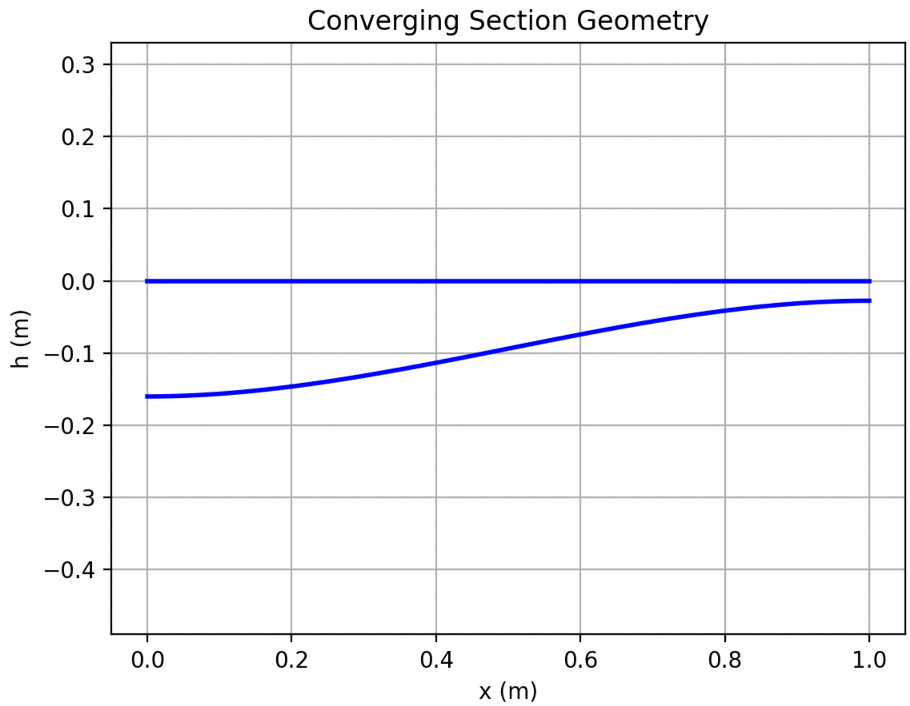

Figure 4: Example of Converging Section Geometry (Note: This is not the final design)

The flow properties along the section were calculated by, first, iteratively solving for M using the area-Mach relation, shown in Equation 2, and taking the subsonic solution. Then, with M at every point and assuming the stagnation properties are conserved, the pressure and temperature at every point is calculated using the isentropic relations, shown in Equations 1a and 1b.

Analysis

For analysis, the calculations are identical to the design phase but instead of generating a contour, a fixed geometry is input. This creates some conflicts, however, because if the input Mach is off design, the calculated A* value does not align with the throat of the designed converging section. If the input Mach is lower than designed for, the calculated A* is smaller than the designed throat, leading to subsonic flow in the nozzle. If the input Mach is higher than designed for, the calculated A* is larger than the designed throat and the flow chokes prematurely. However, the designed geometry continues converging even after the A* position, which creates an impossible flow situation. According to the differential form of the area-Mach relation, Equation 8,

$$\frac{dM}{dx} = \frac{M}{A(M^2 - 1)} \cdot \frac{dA}{dx} \tag{8}$$

the flow cannot stay sonic because if M = 1 and dA/dx < 0, dM/dx approaches infinity, so the Mach cannot stay at 1, it must change. For Mach to remain at 1, dA/dx must equal 0, which is the throat. The flow also cannot go supersonic smoothly because dM/dx approaches infinity at M=1, so no continuous isentropic solution exists. Therefore, our quasi-1D isentropic flow assumption breaks down. In reality, pressure builds up in the converging section and propagates back upstream.

Nozzle

Design

The purpose of a nozzle of a ramjet is to expand the sonic flow from the converging section to supersonic and generate thrust as a result. During the design phase of the project, the minimum length nozzle contour was generated using the 2D Method of Characteristics (MOC) and left variable such that uniform flow and target exit pressure is achieved at the nozzle exit regardless of the input conditions. The target exit pressure is set to the ambient atmospheric pressure to achieve perfect expansion, maximizing exit velocity and avoiding any pressure forces opposing the thrust.

The stagnation properties at the inlet, also the throat, were calculated using the isentropic relations shown in Equations 1a and 1b. These properties are assumed to remain constant throughout the nozzle as the flow is assumed to be inviscid, adiabatic, and steady. The exit Mach number was determined from the target exit pressure using the isentropic pressure-Mach relation,

$$M_{exit} = \sqrt{\frac{2}{\gamma - 1}\left(\left(\frac{P_0}{P_{exit}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right)} \tag{9}$$

The MOC design begins at the throat with an expansion fan centered at the corner where the converging section meets the nozzle. The expansion fan consists of a family of C- characteristics that propagate upward and to the right from the throat toward the centerline, spreading from the initial flow angle θ = 0 deg to a max angle of $\theta_{max} = \frac{\nu_{exit}}{2}$, where ν is the Prandtl-Meyer function,

$$\nu(M) = \sqrt{\frac{\gamma + 1}{\gamma - 1}} \cdot arctan\sqrt{\frac{\gamma - 1}{\gamma + 1}(M^2 - 1)} - arctan\sqrt{M^2 - 1} \tag{10}$$

This expansion fan accelerates the flow from sonic (M = 1) at the throat to the target exit Mach number. Each C- characteristic carries constant values of the Riemann invariant $K^- = \theta + \nu$.

As the C- characteristics reach the centerline, they reflect as C+ characteristics due to the symmetry boundary condition (θ = 0 at centerline). At the centerline, where the reflection occurs, $K^+ = -K^-$. These reflected C+ characteristics propagate downward and to the right from the centerline toward the wall, carrying the Riemann invariant $K^+ = \theta - \nu$. Using a layer-by-layer approach, each reflected C+ characteristic first intersects with the subsequent C- characteristics before reaching the wall. For example, the C+ reflected from the first C-characteristic intersects with the second C- characteristic, then the third, and so on, creating a network of intersection points throughout the nozzle. An example of this layer-by-layer approach can be seen in Figure 5, where the numbered nodes are the order of processing.

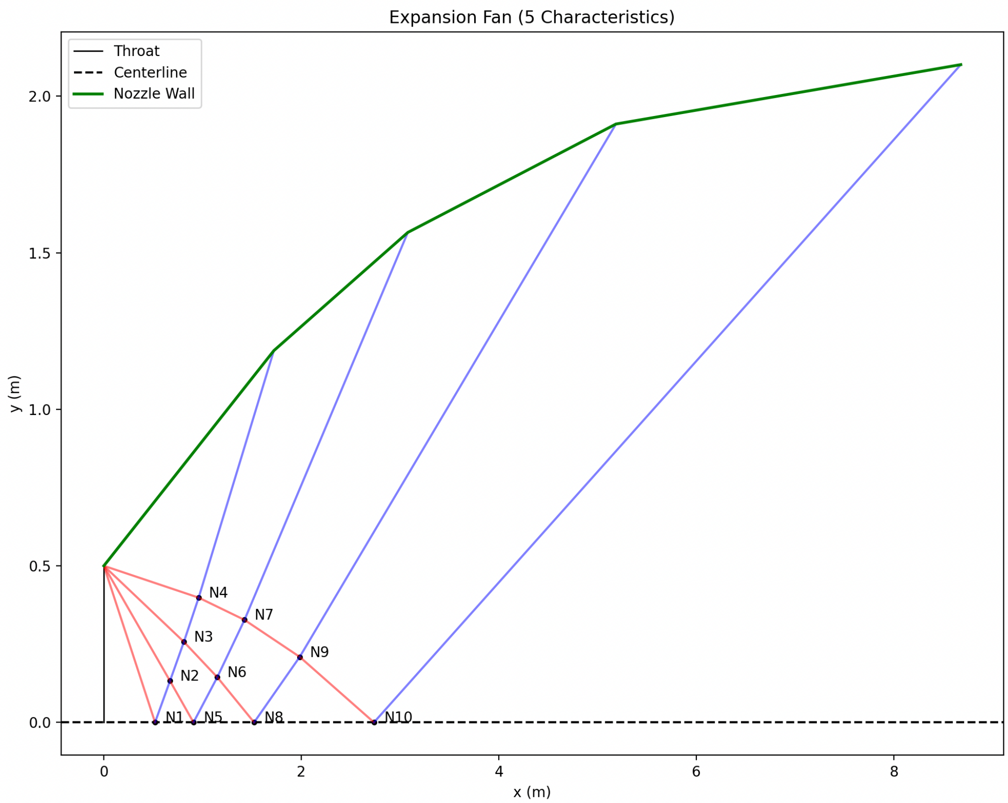

Figure 5: Example of Layer-by-Layer Approach (Note: This is not the final nozzle design)

The intersection points between C- and C+ characteristics form a network throughout the nozzle, and at each intersection, the flow properties are calculated by solving Equations 11 and 12.

$$\theta = \frac{(K^{+} + K^{-})}{2} \tag{11}$$

$$\nu = \frac{(K^{-} - K^{+})}{2} \tag{12}$$

The Mach number at each intersection is then found by inverting the Prandtl-Meyer function (Equation 8) using numerical methods. At each intersection point in the characteristic network, the local flow angle, θ, and Mach angle, $\mu = \arcsin\left(\frac{1}{M}\right)$ combine to determine the directions of the characteristics emanating from that point, where C+ characteristics propagate at $\theta + \mu$ while C- characteristics propagate at $\theta - \mu$.

After the C+ and C- interactions, the nozzle wall contour is formed layer-by-layer as each C+ characteristic propagates from the centerline toward the wall. At each layer, the required wall angle is calculated from the local flow angle, then averaged with the previous wall angle and constrained to pass through the previous wall point. The intersection of the C+ characteristic with this wall segment defines the next wall point. This process is repeated across all characteristics, building up the wall geometry point-by-point. Each C+ characteristic intersects the wall at the junction between adjacent wall panels. As the number of characteristics increases, the wall is divided into more segments, and the averaging process produces finer resolution. As the number of characteristics increases, the nozzle contour converges to a smooth, continuous wall that ensures that the flow at the nozzle exit is parallel to the centerline ($\theta = \theta_{max}$ everywhere), producing uniform, axial flow. An example of the characteristics and resulting nozzle geometry is shown in Figure 6.

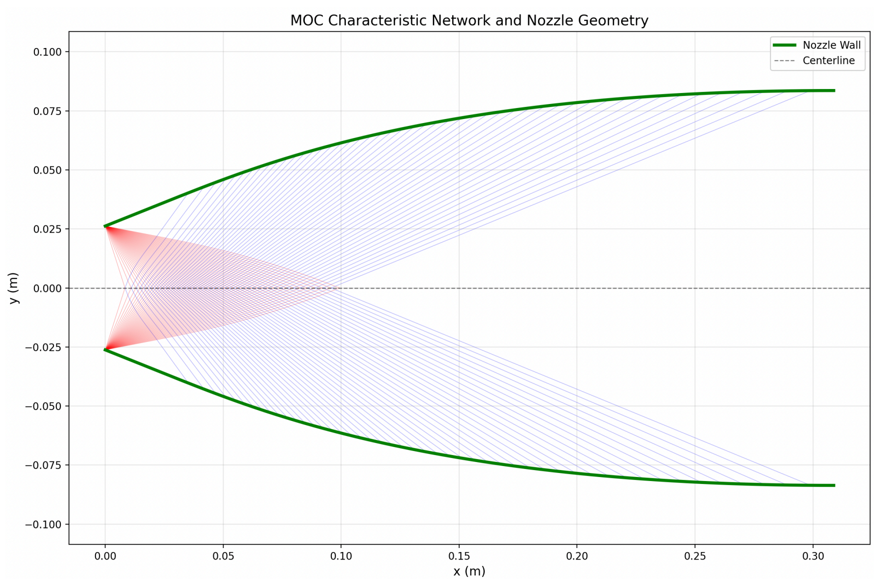

Figure 6: Example of 2D MOC Nozzle Design (Note: This is not the final nozzle design)

Analysis

For analysis, a fixed nozzle geometry generated during the design phase is input. The MOC algorithm generates the same characteristic network as in the design phase: C-characteristics emanate from the throat and propagate upward and to the right toward the centerline, where they reflect as C+ characteristics that propagate downward and to the right toward the wall. However, instead of allowing the C+ characteristics to define the wall contour, they are extended until they intersect the specified wall geometry. The exit Mach number is estimated from the area ratio $\frac{A_{exit}}{A_{throat}}$ using the area-Mach relation (Equation 2), and this determines the required expansion fan angle. The flow properties are then calculated at each characteristic intersection as in the design phase.

To obtain quasi-1D profiles for integration with other ramjet components, the two-dimensional flow field from MOC is reduced to one dimension using mass-flux weighted averaging. At each location x across the nozzle, the flow properties are averaged across the nozzle height weighted by the local mass flux ρu:

$$\bar{M}(x) = \frac{\int_0^{h(x)} M(x,y)\rho(x,y)u(x,y)dy}{\int_0^{h(x)} \rho(x,y)u(x,y)dy} \tag{13}$$

The static pressure and temperature are then calculated from the averaged Mach number using the isentropic relations to ensure that the stagnation properties are conserved.

The behavior of flow in the nozzle during operation is highly dependent on the nozzle inlet conditions. The nozzle is designed assuming sonic flow at the throat. If the inlet Mach number is somehow supersonic, the local flow angle will not equal the wall angle, leading to C+ characteristics not straightening out and producing non-uniform flow at the exit. If the inlet Mach number is subsonic, no expansion fan will form because expansion fans only occur in supersonic flow. Instead, the diverging geometry will cause the subsonic flow to decelerate further, leading to continued pressure rise and velocity decrease. Also, off-design exit pressure ratios would cause expansion waves or oblique shocks to form at the nozzle exit to match the ambient pressure.

Final Design

Profile

All engine section design functions were linked sequentially, with each component designed in order. The inlet is designed first, and its exit flow conditions are passed directly to the diffuser design module. The diffuser exit conditions are then used as inputs for the flameholder design, and this process continues downstream for each subsequent engine section.

The engine was designed for operation at an altitude of 55,000 ft. At this altitude, free-stream conditions of $P_\infty = 9112.32$ Pa and $T_\infty = 216.65$ K were obtained from the 1976 U.S. Standard Atmosphere model. As noted previously, the inlet design is constrained by the maximum operating Mach number of $M = 3.25$; therefore, the entire engine was designed to operate at this Mach number.

Each engine section has its own design parameters, which we list in Table 2.

Table 2: Final Design Parameters

	Parameter	Value	Units
General	Incoming Mach number	3.25	-
	Air mass flow rate	10	kg/s
	Depth into-the-page	1	m
Inlet	Turn angles	10, 10, 10	Degrees
Diffuser	Exit Mach number	0.1	-
	Length	0.1	M
Flameholder	-	-	-
Combustor	External Mach number	2.75	-
	Fuel mass flow rate	0.1	kg/s
Converging Section	Length	0.1	M

It should be noted that for the combustor, we *must* design for the lowest Mach regime, in order to completely combust all fuel to fully add required heat, when flying off-design. However, the inlet is designed for the highest Mach flow. To reconcile this when designing the combustor, we first design all components up to- (but not including) the combustor, then run an auxiliary analysis script that outputs the Mach, pressure, and temperature values. These are the values that are then used for the combustor design.

When these parameters are used to design their respective components, a profile of the inner surface of the ramjet can be found. However, this is not enough for a full ramjet design because the walls must have some thickness. In addition, the inner profile should be encapsulated in some more aerodynamic shell to minimize pressure drag. To do that, we surrounded the inner profile with outer walls to fully contain the entire profile. This forms our final ramjet profile and can be seen in Figure 7.

This outer profile also adds two surfaces which contribute to pressure drag- the outer surface of the lip and the lower surface of the inlet. The design of the outer surface of the lip was described earlier, and the pressure calculations follow from that design- if the jet is operating below its maximum Mach number, the incoming flow over the lip is parallel to the lip and has the same conditions as the inlet flow after all oblique shocks, before the final normal shock. If the jet is operating at its maximum Mach number, the incoming flow over the lip is horizontal and has atmospheric conditions, so an oblique shock will be formed to turn it along the lip. In both of these situations, we can use the oblique shock equations to calculate the pressure force on the lip.

The design of the outer surface of the bottom profile was made such that the surface between the end of the diffuser and the end of the nozzle would be horizontal. This was done to reduce the amount of pressure drag along that surface, while the region under the inlet and diffuser is a straight line traveling from tip of the inlet to the end of the horizontal line. The lower surface of the inlet is comparatively simpler to analyze- for all operating Mach numbers, it experiences horizontal flow at atmospheric conditions. An oblique shock will form to turn the flow along the surface, so the pressure on the surface can be found using oblique shock formulas.

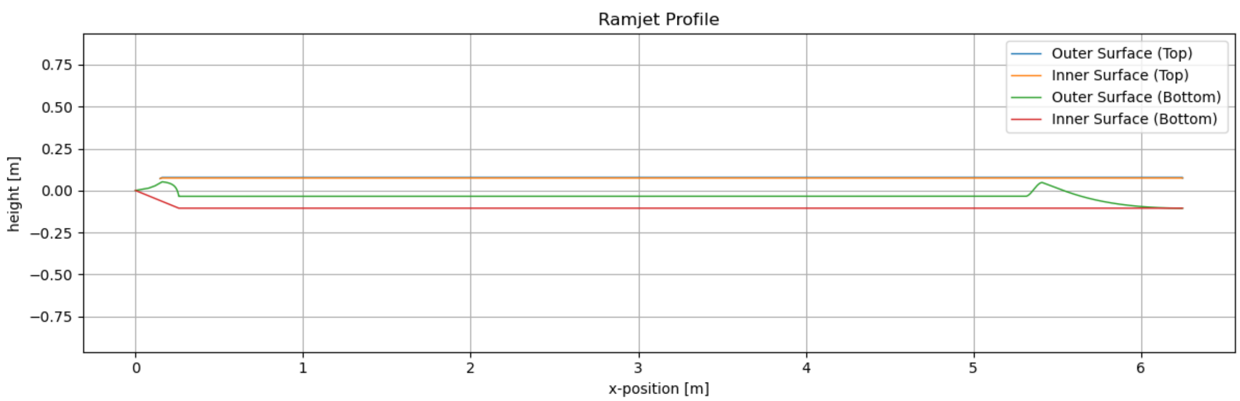

Figure 7: Ramjet full profile.

Control volume

As mentioned earlier, to calculate the thrust of the ramjet design, a control volume analysis was performed. A control surface was designated following the external surface of the ramjet wherever possible. There are two points where this causes issues: the back of the inlet and the end of the nozzle. At the inlet, the control surface follows the lower segmented surface until it reaches the point closest to the lip. Because this is the closest point on the surface to the lip and the air flow over the surface is parallel to the surface, the control surface can then go from the surface to the lip orthogonal to the flow through the inlet. Once this is done, the control surface can follow the top surface of the ramjet to the back of the design until the end of the nozzle. Then, because the flow at the rear of the nozzle is uniform horizontal flow, the control surface can go directly orthogonally across this flow until it reaches the bottom of the nozzle, at which point it can continue along the bottom surface of the ramjet to meet its starting point. A full diagram of the control surface, shown as the red dotted line, can be found in Figure 8.

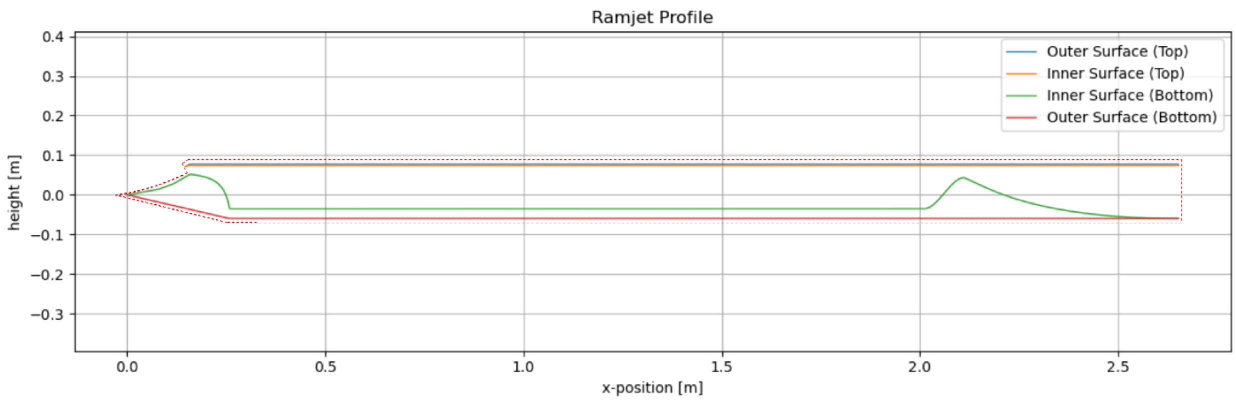

Figure 8: Ramjet Control Volume (Note: Not to scale. For visual purposes. In the analysis, the dotted line is hugging the ramjet.).

The reasoning for this specific control surface is that it simplifies analysis. At every point along the control surface, both pressure forces and momentum flux in the x-direction must be considered. By making the control surface parallel to the x-axis as much as possible, we can neglect pressure forces along much of the surface. In addition, by making the control surface orthogonal to the flow wherever it crosses flow, we do not have to consider the angle of the flow relative to the surface, which simplifies the necessary calculations.

Results

Operation at different speeds

Firstly, after designing the full profile, we checked that all our fuel could combust in the combustor, via the plot displayed in Figure 9: Burn time of fuel and flow time of air through combustor, versus operating Mach number.We confirmed that all fuel, in fact, does burn, as the burn time is always equal to or lower than the time spent within the combustor (flow time).

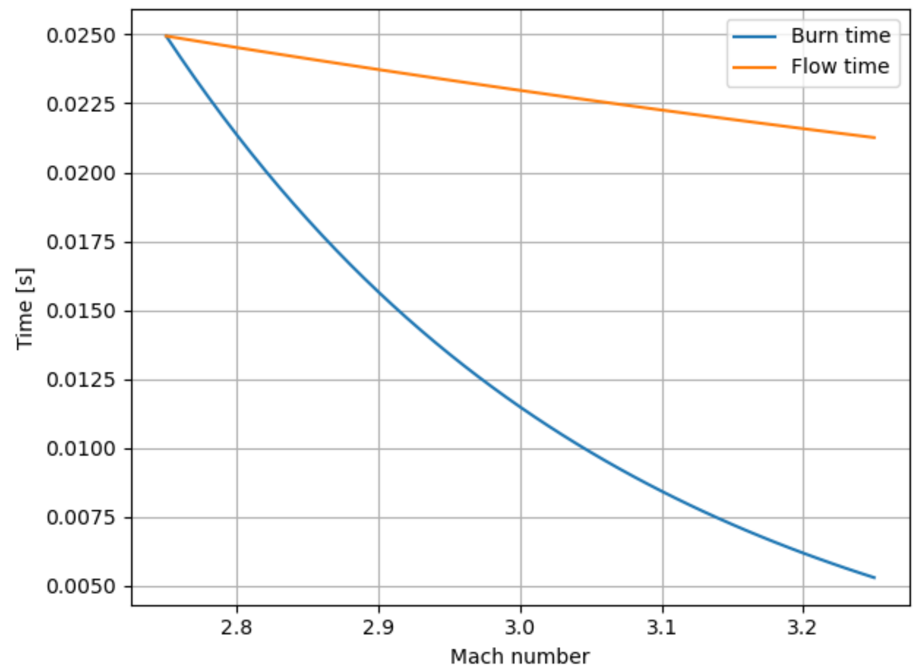

Figure 9: Burn time of fuel and flow time of air through combustor, versus operating Mach number.

After performing control volume analysis, we can find the overall thrust produced by the ramjet. This thrust, however, depends on the speed of the incoming air because incoming airspeed changes all properties of the flow within the engine, which effects the output speed. Thus, we can create a graph which shows the thrust of the engine at every speed in its operating range, as seen in Figure 10. The thrust is positive throughout the entire operating range, going from 1500 newtons at Mach 2.75 to 5000 newtons at Mach 3.25.

In addition, the fuel flow required to achieve these thrusts changes depending on the Mach number of the surrounding air. This is because the amount of fuel that needs to be injected is constrained by the requirement that the flow must choke at the throat of the converging section. If the input conditions to the converging section is off design, the flow will either never reach sonic by the time it reaches the throat, leading to subsonic flow in the nozzle, or the flow will try to choke early, leading to quasi-1D assumption breaking down. Thus, for different oncoming Mach numbers, different fuel flow rates are required to ensure the input properties into the converging section is on design. The results of this analysis can be seen in Figure 11.

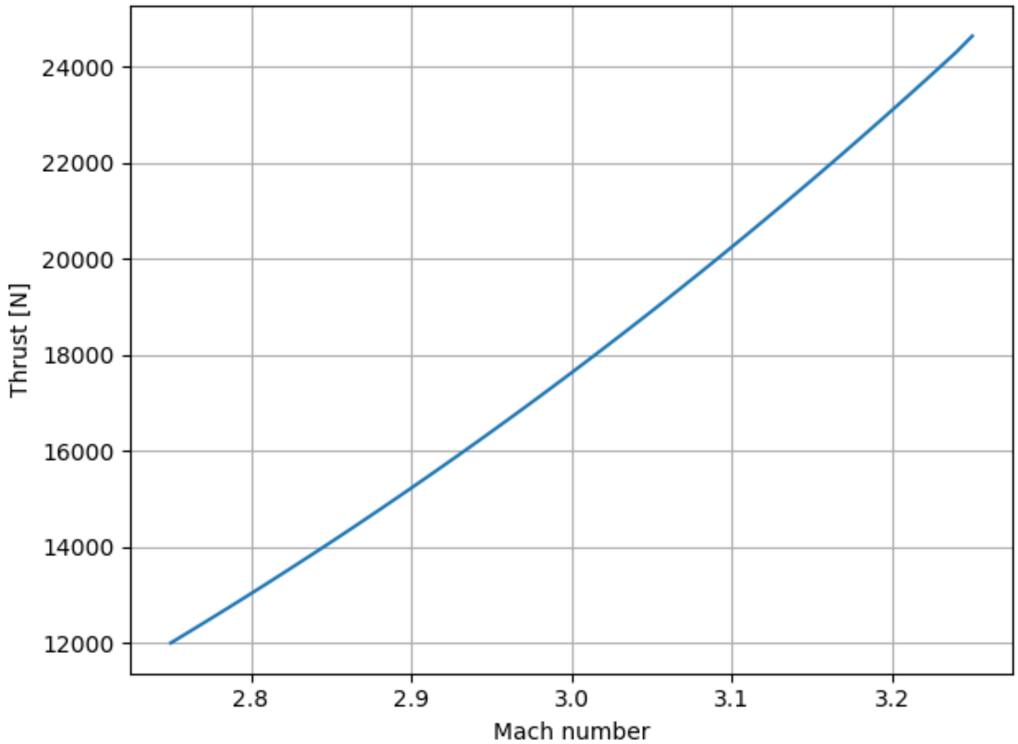

Figure 10: Thrust vs Mach number.

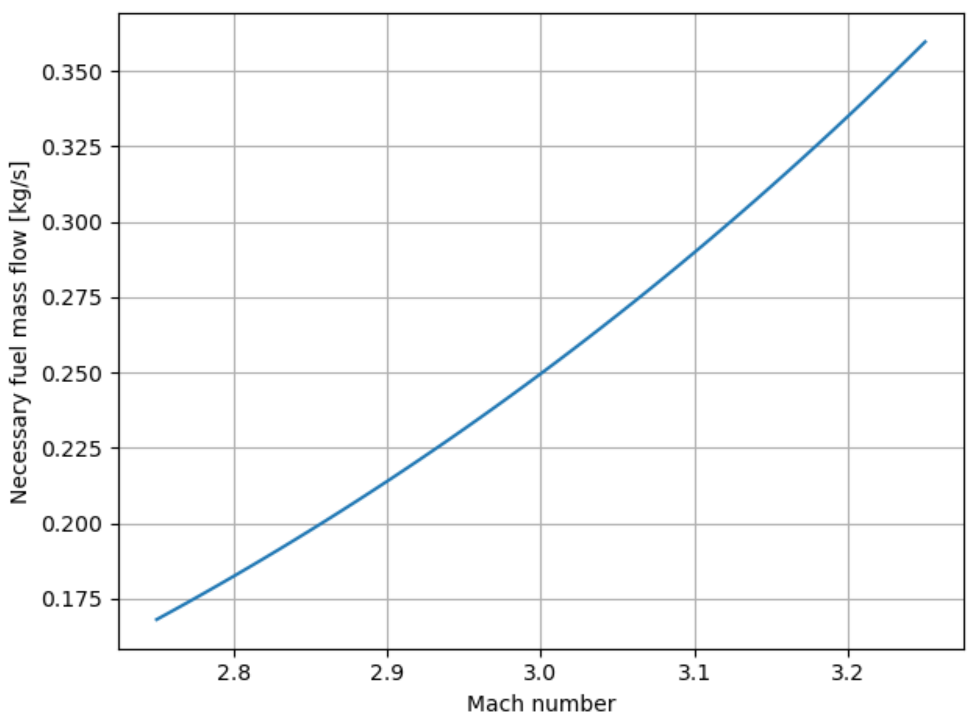

Figure 11: Fuel flow vs Mach number.

Parameter distributions

In addition to the overall jet thrust calculations, we found property profiles over the length of the jet. The diffuser, flameholder, combustor, and converging section were all modeled as 1-dimensional sections, so 1-dimensional property distributions arise naturally from the analysis. The inlet and the nozzle, on the other hand, were designed using 2-dimensional methods, which naturally produce 2-dimensional property distributions. In order to get 1-dimensional distributions, we averaged properties by mass flux along each vertical slice of the component to produce 1-dimensional profiles.

The properties for which we found profiles were Mach number, entropy relative to input, static pressure, stagnation pressure, static temperature, and stagnation temperature. These can be seen in Figures 12 through 17, respectively. These distributions all match expected behavior, and each component changes the flow in the expected manner.

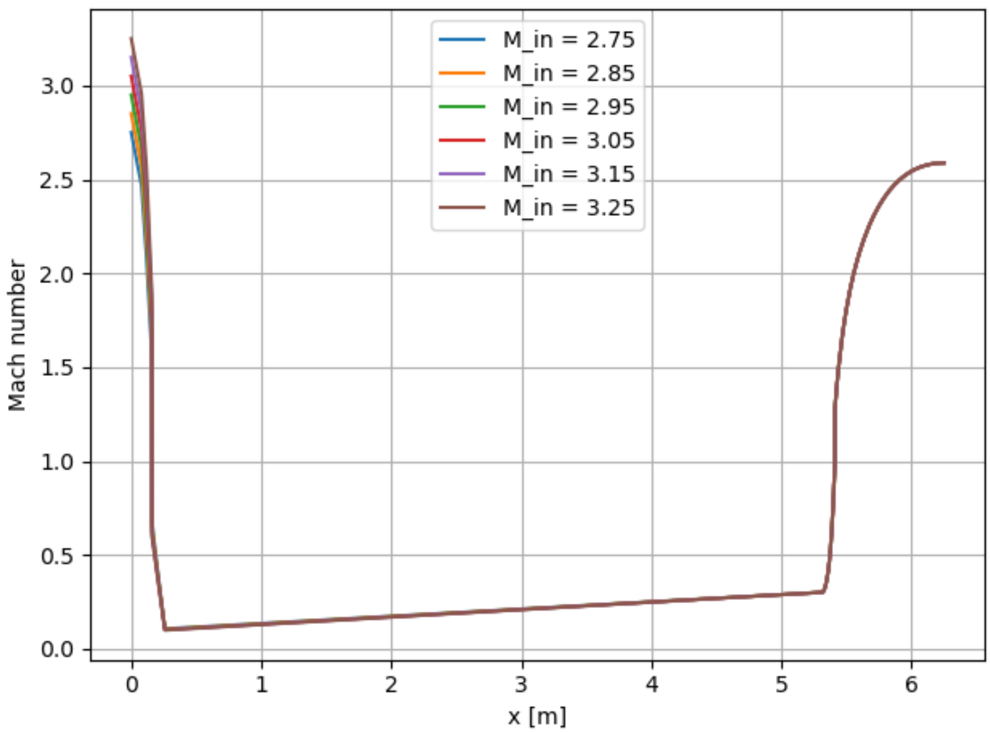

Figure 12: Mach number profile

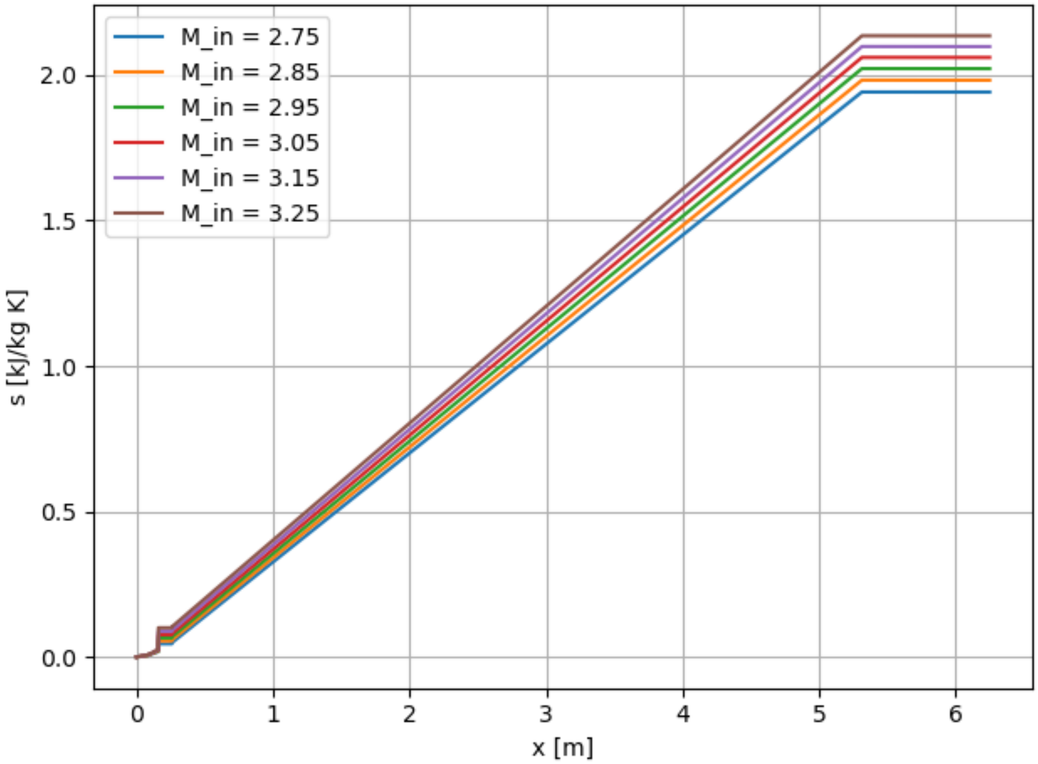

Figure 13: Entropy profile.

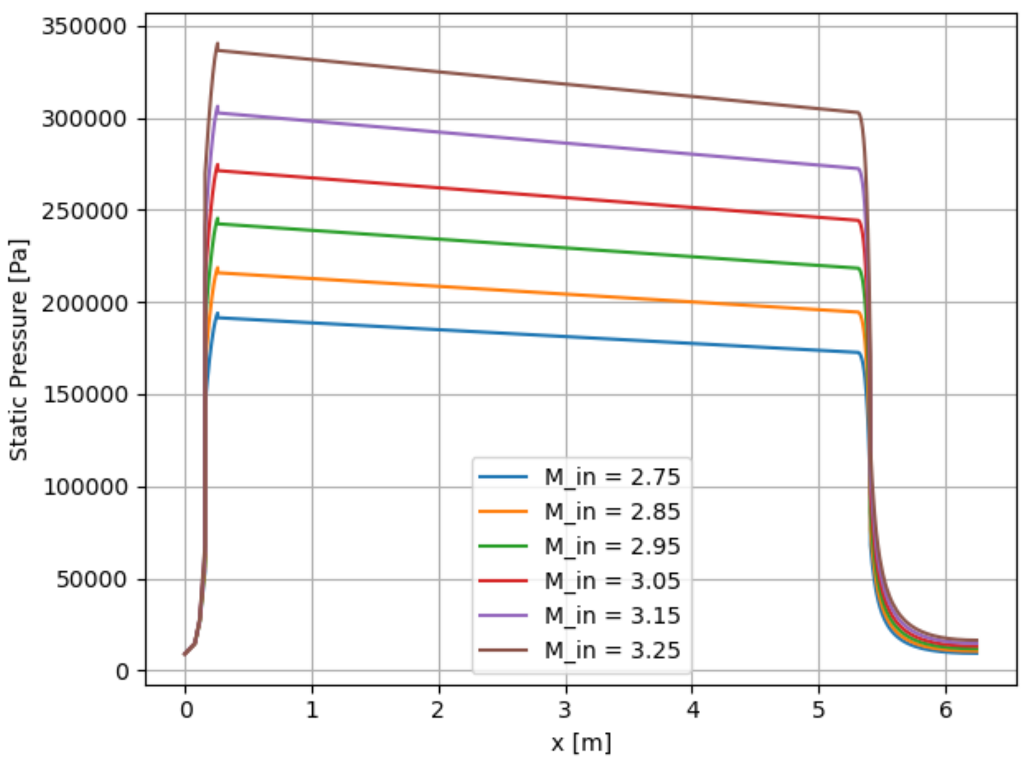

Figure 14: Static pressure profile.

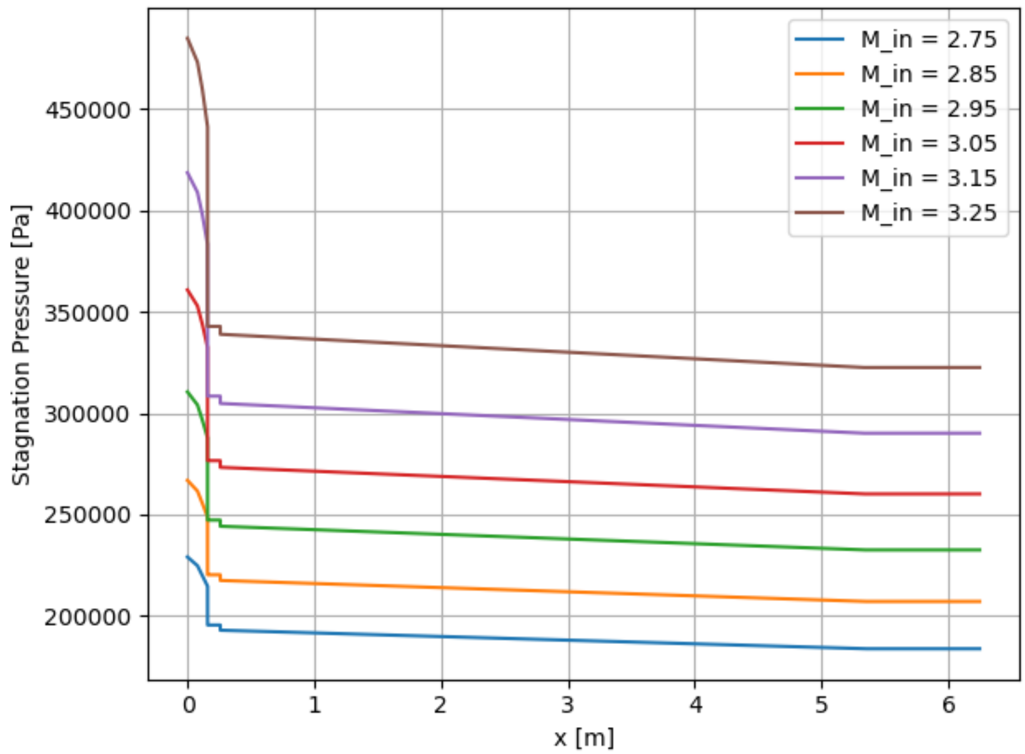

Figure 15: Stagnation pressure profile.

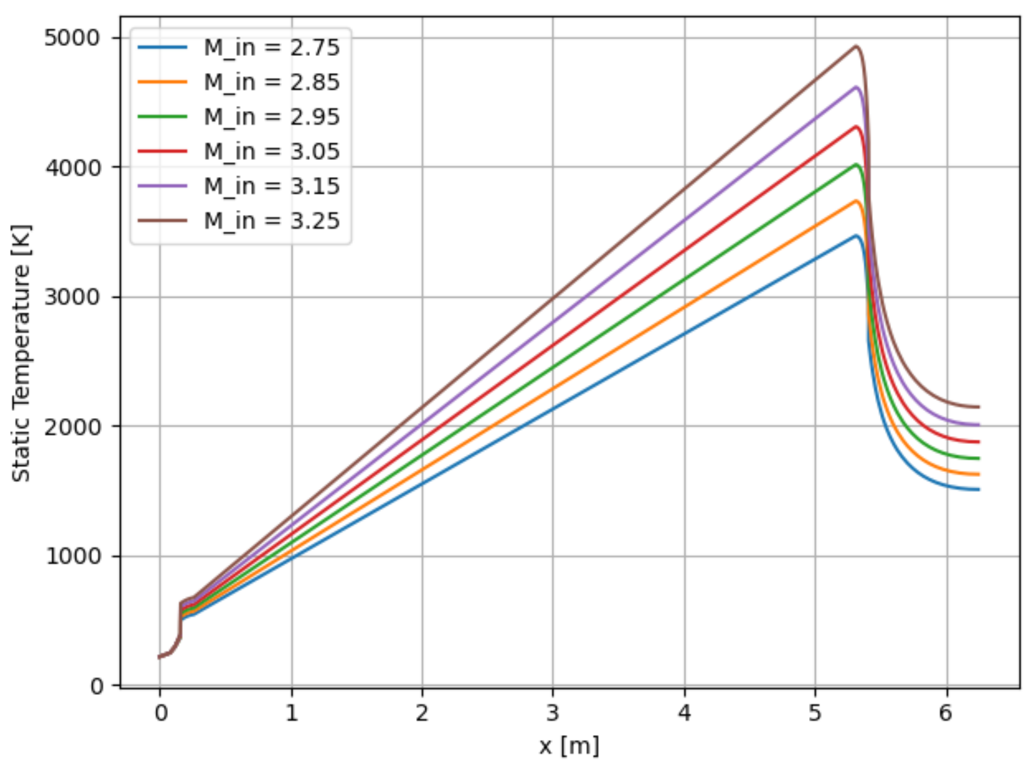

Figure 16: Static temperature profile.

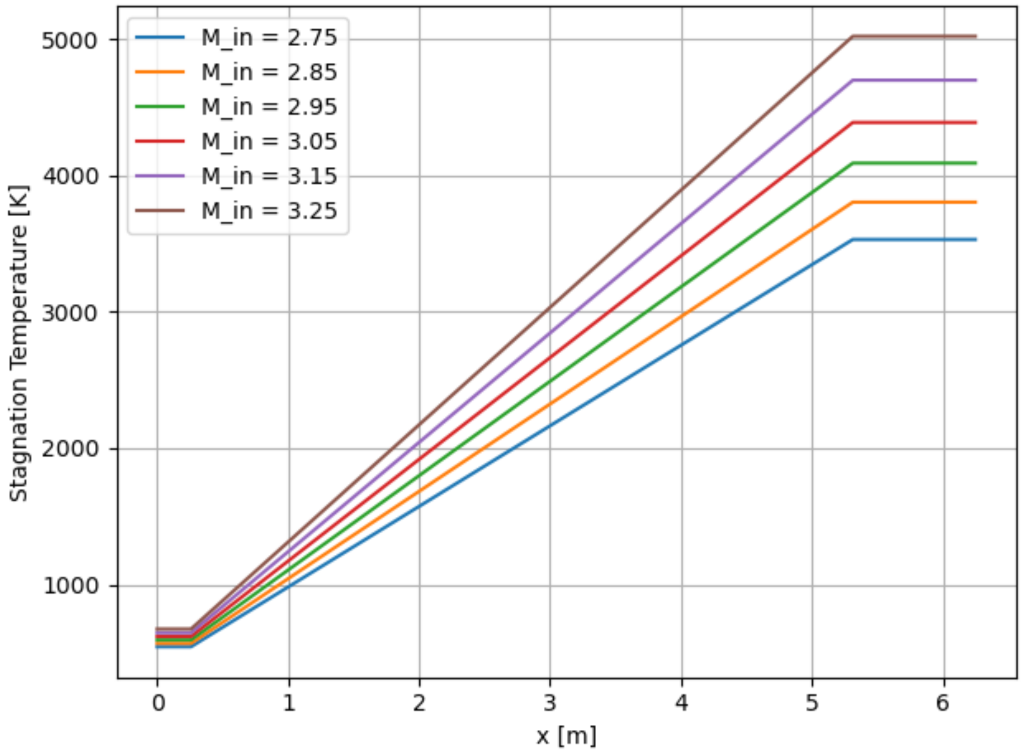

Figure 17: Stagnation temperature profile.

Thermodynamic analysis

Given the parameter distributions we produced previously, we were also able to perform a thermodynamic analysis of the Brayton cycle which powers the ramjet. To do this, we made P-v and T-s diagrams, as shown in Figure 18 and Figure 19, respectively. These diagrams look like the expected diagram shapes for a Brayton cycle. In addition, it is clear in both diagrams that at higher Mach numbers, the pressure ratio of the cycle is higher, leading to increased efficiency. This indicates that the ramjet consumes fuel more efficiently at higher Mach numbers. This analysis could be used to perform a numerical first or second law efficiency analysis, but that was out of the scope of this project.

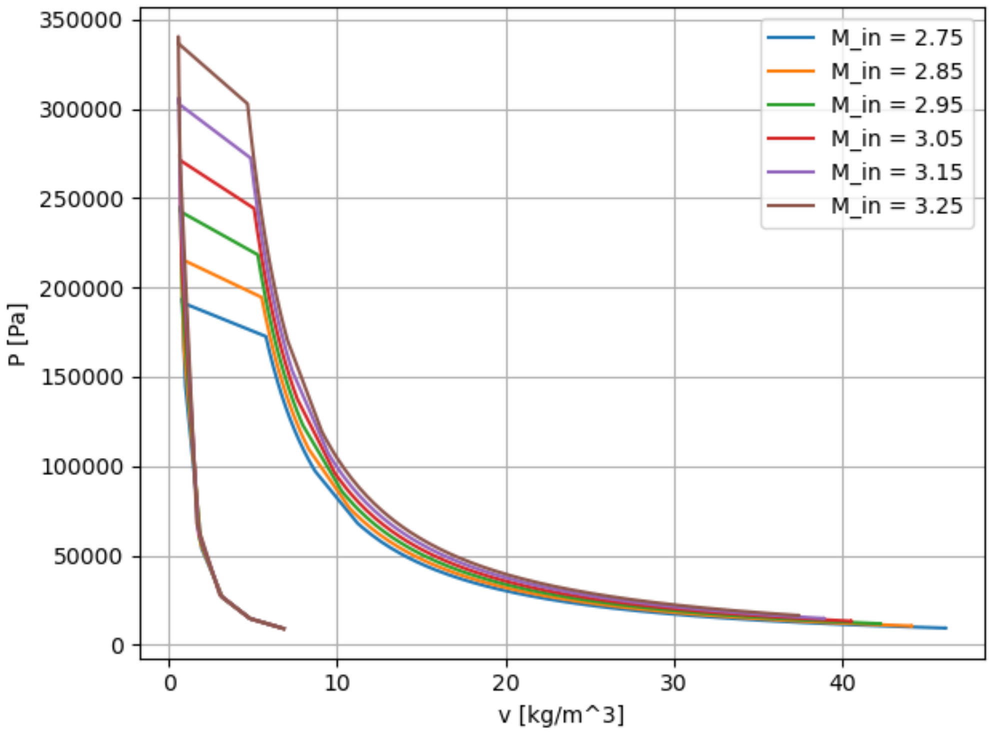

Figure 18: P-v diagram.

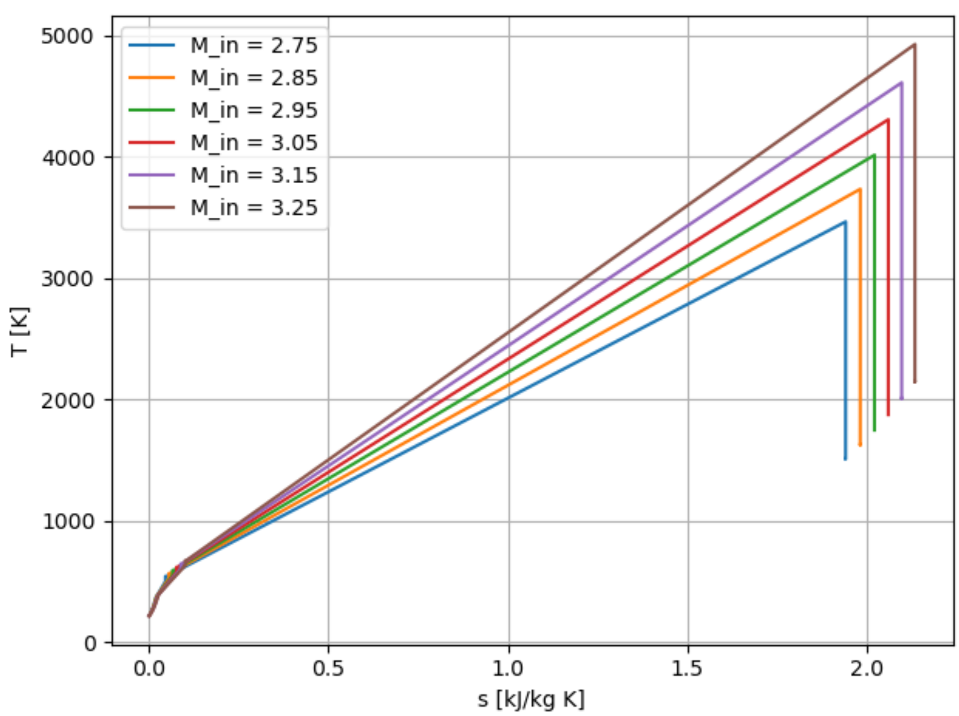

Figure 19: T-s diagram

Conclusion and Future Work

In this project, we successfully designed a ramjet. This ramjet produces 1582.25 - 5028.21 newtons of thrust at atmospheric conditions at 55,000 ft, operating between Mach 2.75 and Mach 3.25.

There are multiple possible improvements which could be made to this project. One major possible improvement is relaxing many of the assumptions we made. Specifically, we assumed that the fluid entirely consisted of air, which was modeled calorically perfect ideal gas. The fluid consisting entirely of air was a reasonable assumption, as fuel is added on the order of $\sim 0.1\ kg/s$, whereas the air flow rate was on the order of $\sim 10\ kg/s$, but even this small (~1%) change in fluid composition could lead to changes in properties which should be accounted for. In addition, the calorically perfect assumption is reasonable at low temperatures, but starts to break down at higher temperatures. The highest temperature reached by this ramjet is ~1250 K, which is high enough to lead to changes in c_v and c_p which should be accounted for by relaxing the calorically perfect assumption.

In addition, we ignored viscous drag in our analysis, and relaxing this assumption could lead to different design choices. Specifically, horizontal faces like the top and bottom faces of the jet or near-horizontal faces like the forwardmost faces of the inlet contributed no drag in our analysis, whereas in reality they would contribute significant viscous drag. Accounting for viscous drag would penalize these parts of our designs more, possibly leading us to alternative designs.

Finally, future work could consist of optimizing this design under this or any different set of assumptions. For this project, once we found a jet design that functioned and produced positive thrust, we did not at all attempt to optimize. It may be valuable to optimize this design for some combination of parameters, which could be thrust but could also incorporate some combination of thrust, fuel consumption, length, and size. Mass could also be considered, although this would require calculating engine mass, which we did not do.